

Hotel Bar Inventory Forecasting

Executive summary

The system converts transaction-level bottle balances into daily demand by bar and brand, forecasts future consumption, and recommends par levels that balance service availability with holding stock.

Dataset evidence

The source contains 6,575 transaction rows, 6 bars, and 16 brands from 2023-01-01 through 2024-01-01. The maximum conservation error is 0.00000000 ml. The completed daily grid contains 35,136 rows across 96 observed bar-brand pairs.

Forecast evidence

The chronological validation compares a 7-day seasonal-naïve baseline with a Random Forest using lags, rolling statistics, weekday features, bar, and brand. Baseline WAPE: 175.0%. Random Forest WAPE: 176.7%.

Inventory policy

Par level equals two-day lead-time demand plus 95% service-level safety stock. Across the validation policy simulation, stockout days total 418, lost volume totals 68,193 ml, and average inventory totals 33,579 ml across series. The example highest-volume validation series is Thomas's Bar / Yellow Tail. Its simulation records 14 stockout days and 2,400 ml lost volume.

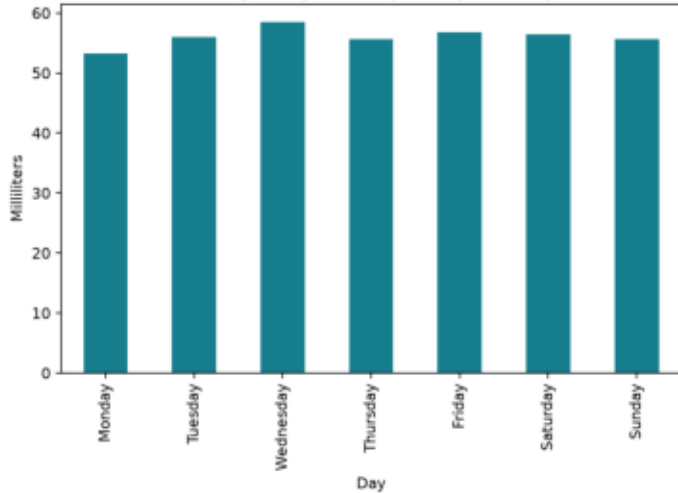
Assumptions and risks

Lead time is constant at two days, unmet demand is treated as lost, and measurements are in milliliters. Historical stockouts may censor observed demand. Promotions, holidays, vendor delays, breakage, and pouring waste should be monitored before production deployment.

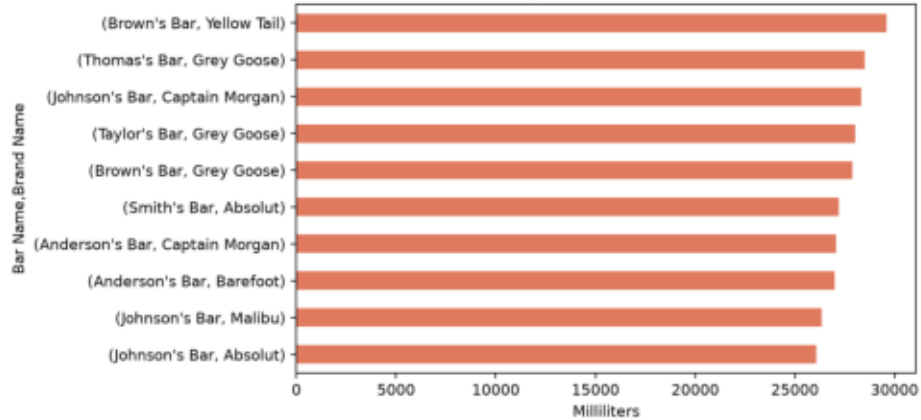
Recommendation

Use the generated recommendation CSV as a daily manager review queue. Prioritize Class A items, confirm vendor lead times, and monitor WAPE, stockouts, lost volume, inventory, and data-quality errors.

Average daily consumption by weekday



Top bar-brand series



Validation inventory path: Thomas's Bar / Yellow Tail

